

Active Directory & Group Policy Implementation Lab

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May 2, 2026

Objective

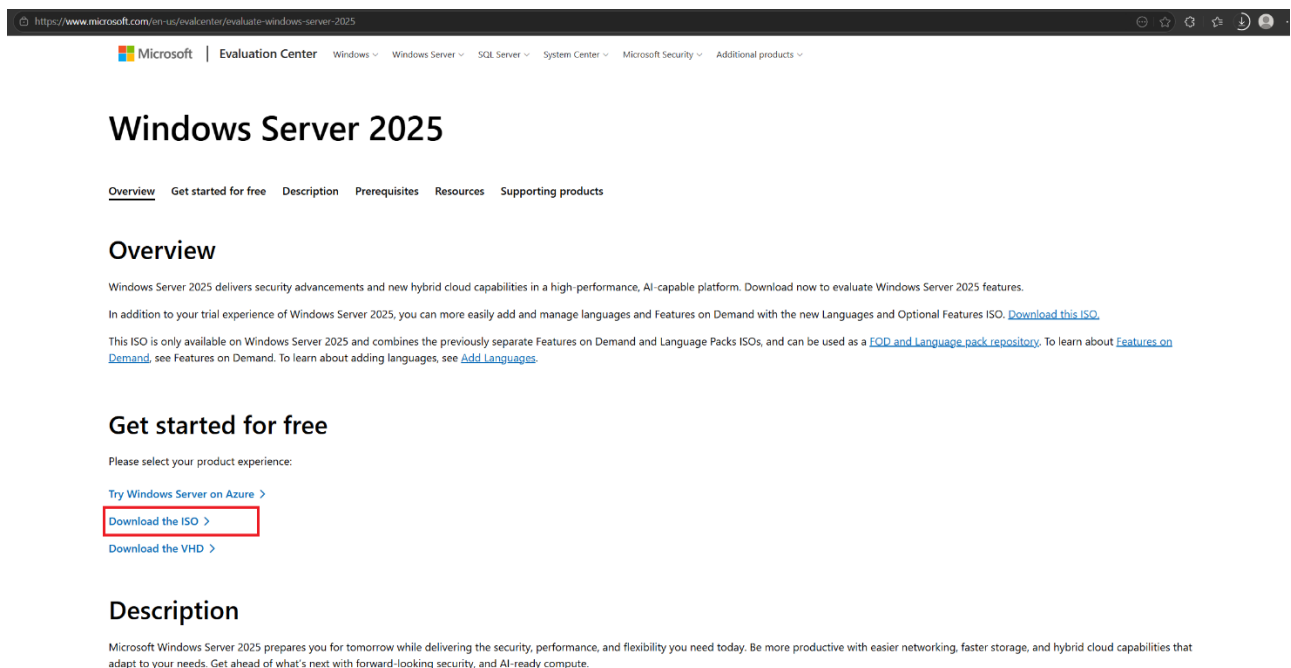
To build a functional Active Directory domain in a virtual environment, create users and organizational units, configure Group Policy for password security, and successfully join and test a client machine.

Technologies & Tools

- VMware Workstation
- Windows Server 2025 Evaluation
- Windows 10 Client
- Active Directory Domain Services (AD DS)
- Group Policy Management Console

1. Windows Server Installation & Active Directory Setup

Installed Windows Server 2025 on a VM (DC01).



The screenshot shows the Microsoft Evaluation Center page for Windows Server 2025. The page has a dark header with the Microsoft logo and navigation links: Evaluation Center, Windows, Windows Server, SQL Server, System Center, Microsoft Security, and Additional products. The main heading is "Windows Server 2025". Below it, there are tabs: Overview (selected), Get started for free, Description, Prerequisites, Resources, and Supporting products. The "Overview" section contains text about the security advancements and hybrid cloud capabilities of Windows Server 2025, and links to download the ISO and VHD. The "Get started for free" section has a link to "Try Windows Server on Azure" and a red-bordered button labeled "Download the ISO". The "Description" section mentions the security, performance, and flexibility of Windows Server 2025.

Microsoft | Evaluation Center Windows Windows Server SQL Server System Center Microsoft Security Additional products

Windows Server 2025

[Overview](#) [Get started for free](#) [Description](#) [Prerequisites](#) [Resources](#) [Supporting products](#)

Overview

Windows Server 2025 delivers security advancements and new hybrid cloud capabilities in a high-performance, AI-capable platform. Download now to evaluate Windows Server 2025 features.

In addition to your trial experience of Windows Server 2025, you can more easily add and manage languages and Features on Demand with the new Languages and Optional Features ISO. [Download this ISO.](#)

This ISO is only available on Windows Server 2025 and combines the previously separate Features on Demand and Language Packs ISOs, and can be used as a [FOD and Language pack repository](#). To learn about [Features on Demand](#), see [Features on Demand](#). To learn about adding languages, see [Add Languages](#).

Get started for free

Please select your product experience:

[Try Windows Server on Azure >](#)

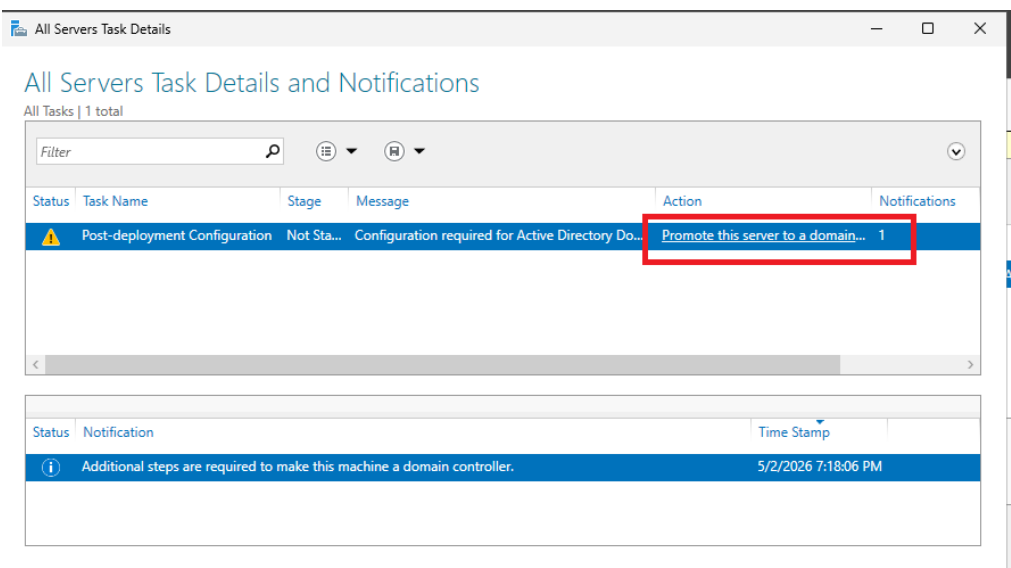
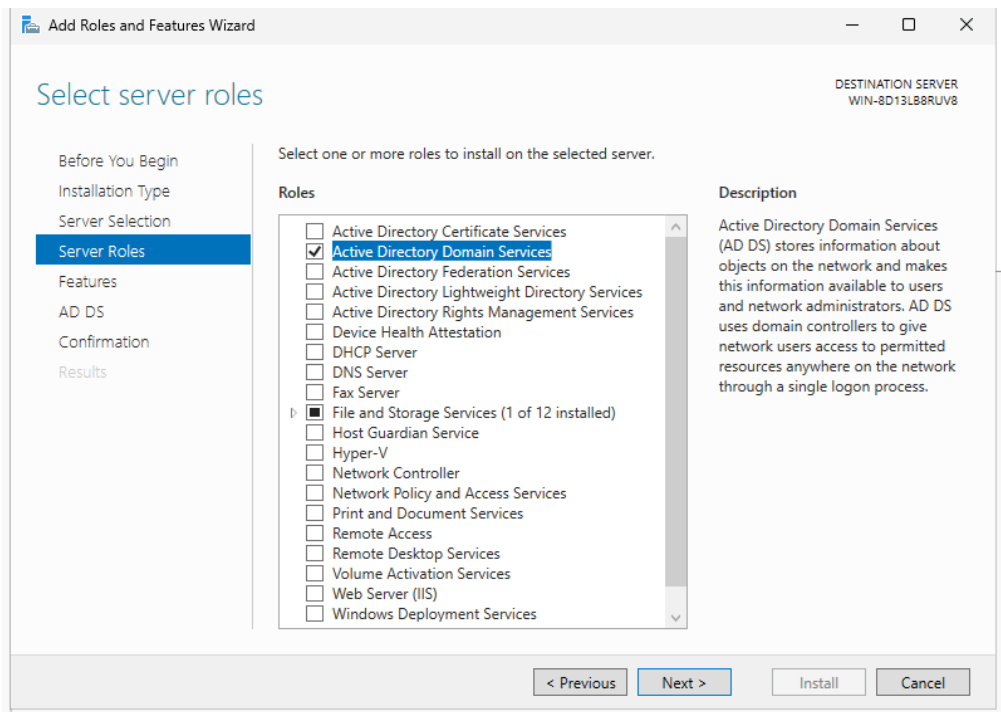
[Download the ISO >](#)

[Download the VHD >](#)

Description

Microsoft Windows Server 2025 prepares you for tomorrow while delivering the security, performance, and flexibility you need today. Be more productive with easier networking, faster storage, and hybrid cloud capabilities that adapt to your needs. Get ahead of what's next with forward-looking security, and AI-ready compute.

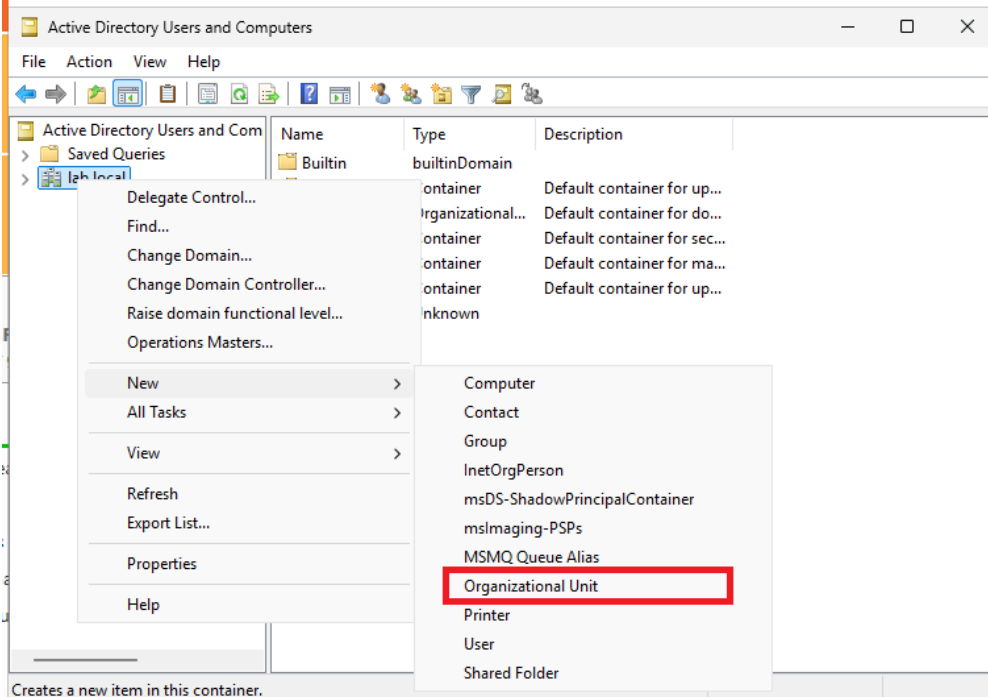
Promoted the server to a Domain Controller and created a new forest with the root domain lab.local.



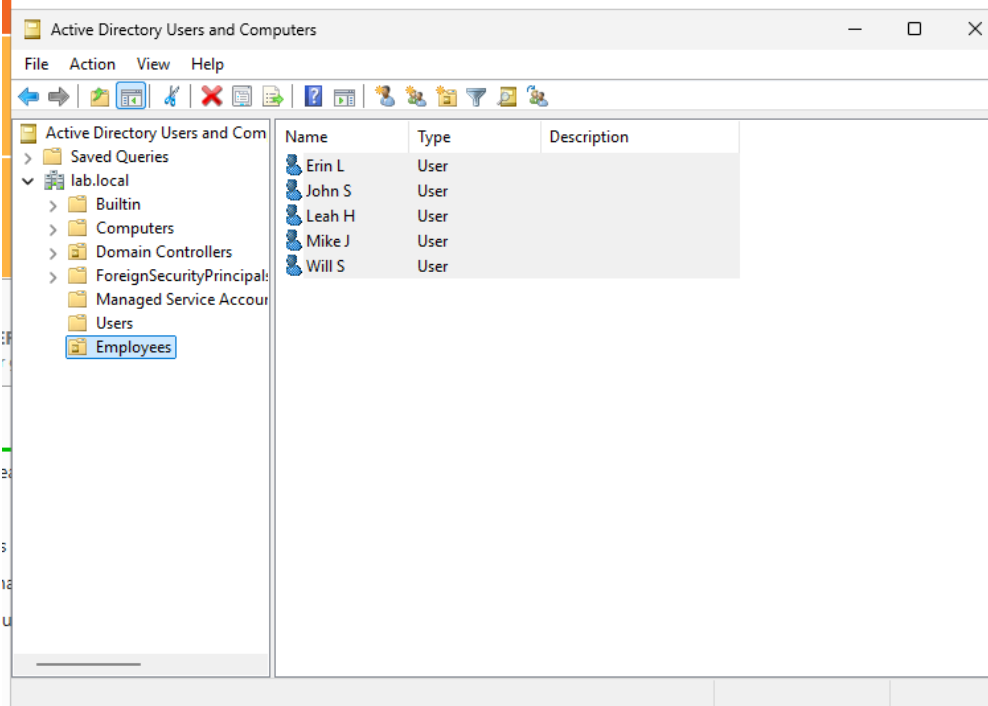
2. Users, Groups, and Organizational Units

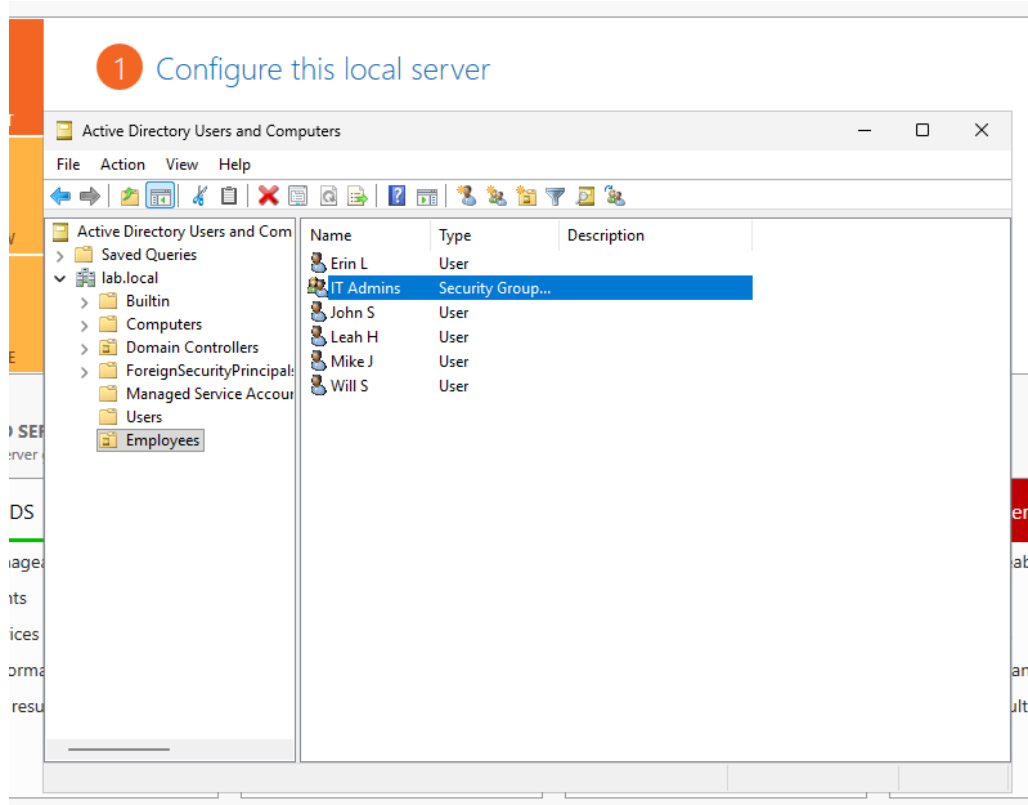
Created an Organizational Unit called **Employees** and added several test users. Also created a security group (**IT_Admins**) and added a user to it.

1 Configure this local server



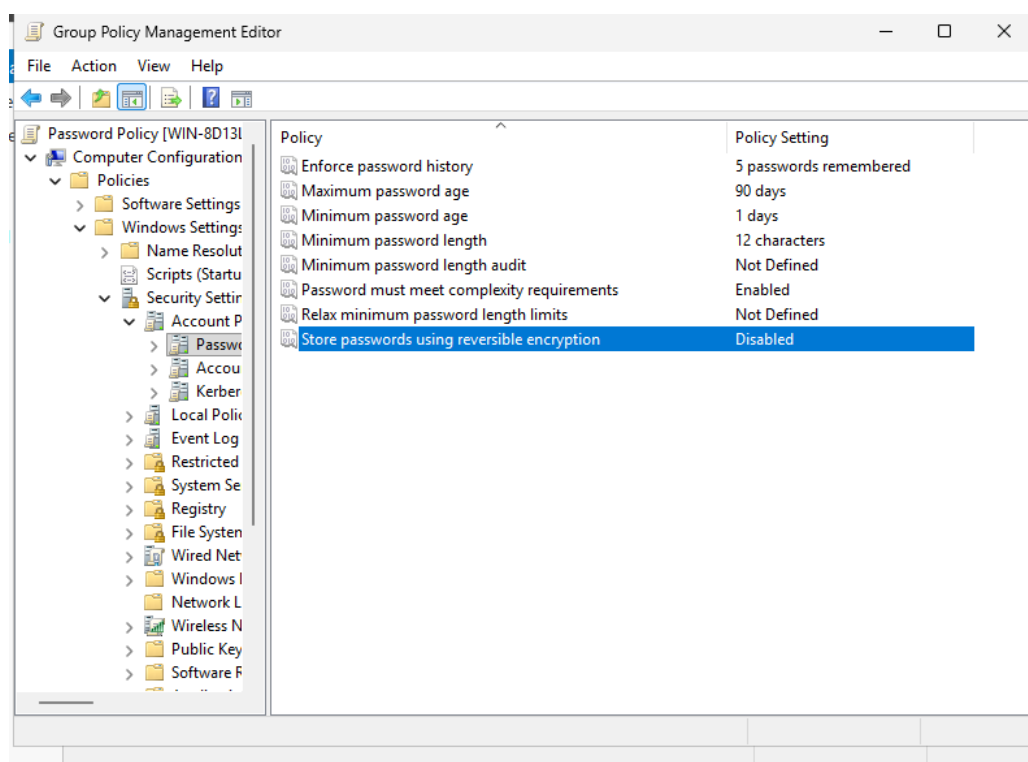
1 Configure this local server





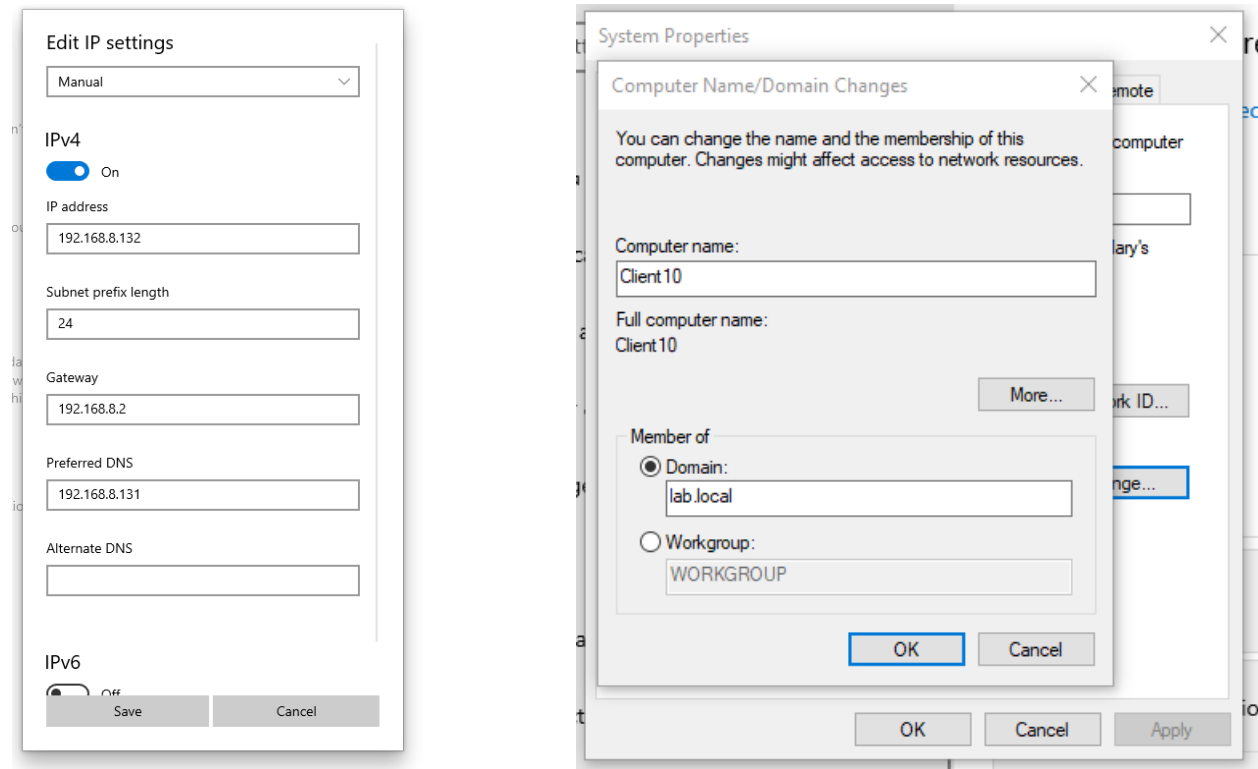
3. Group Policy Configuration

Created a Group Policy Object named **Password Policy** and linked it to the Employees OU. Configured strong password requirements.

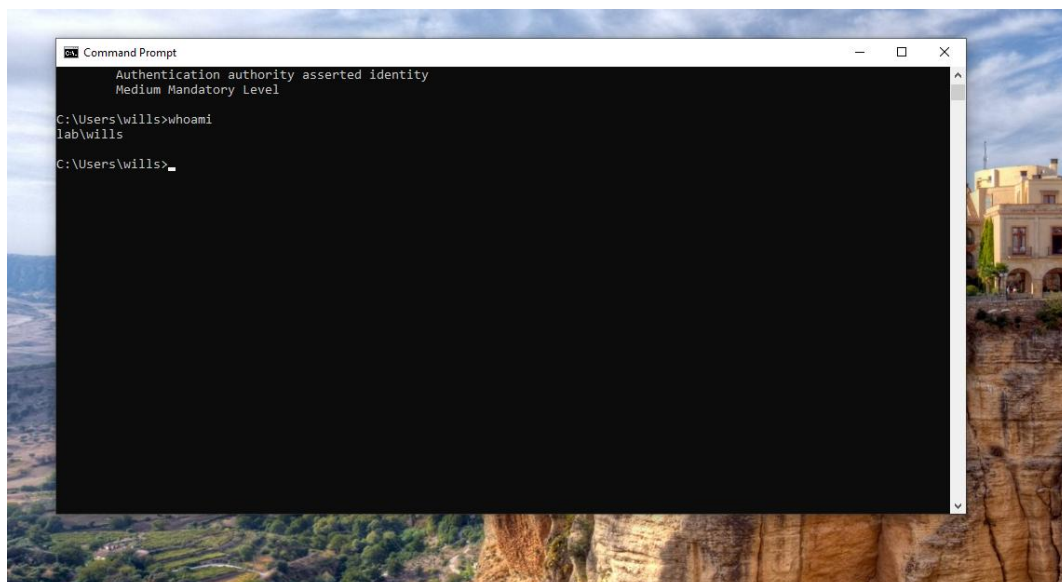


4. Client Configuration and Domain Join

Configured a Windows 10 client VM with static IP addressing in the same subnet as DC01 and pointed DNS to the Domain Controller. Successfully joined the client to the lab.local domain.



Logged into the client as a domain user (LAB\wills). Forced Group Policy update and verified application. Tested password policy enforcement.



```
Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\wills>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on [5/2/2026 at 10:02:01 PM]

RSOP data for LAB\wills on CLIENT10 : Logging Mode
-----
OS Configuration:      Member Workstation
OS Version:            10.0.19045
Site Name:             N/A
Roaming Profile:       N/A
Local Profile:         C:\Users\wills
Connected over a slow link?: No

USER SETTINGS
-----
CN=Will S,OU=Employees,DC=lab,DC=local
Last time Group Policy was applied: 5/2/2026 at 10:00:09 PM
Group Policy was applied from: WIN-8D13LB8RUV8.lab.local
Group Policy slow link threshold: 500 kbps
Domain Name:          LAB
Domain Type:          Windows 2008 or later

Applied Group Policy Objects
-----
N/A

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
Domain Users
Everyone
BUILTIN\Users
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
IT Admins
Authentication authority asserted identity
Medium Mandatory Level

C:\Users\wills>
```

Lessons Learned & Challenges

- Proper DNS configuration is critical for domain communication.
- Group Policy can take time to apply — gpupdate /force and reboots are often required.
- Domain Controllers restrict interactive logons for regular users by default for security.
- Hands-on labs like this build strong foundational skills in enterprise identity and access management.

Conclusion

This lab demonstrates core Active Directory administration and Group Policy management skills essential for IT and Cybersecurity roles.